considered in [6]:

$$\mathcal{H} = -c_\sigma^+ \partial_x^2 c_\sigma + f_\sigma^+ \partial_x^2 f_\sigma - u_0 (n_c - n_f)^2 + \quad (1)$$
$$u_2 \Big(c_\uparrow^+ c_\downarrow^+ f_\downarrow f_\uparrow + h.c. \Big) - k_F^2 (n_c - n_f) - \mu (n_c + n_d)$$

where μ measures a deviation from a perfect nesting. It is worth noticing that the Coulomb interaction, $(n_c + n_f)^2$ term, renormalizes to zero [7]. It is also possible to show that the disparity between electron and hole masses only affects the coefficients in two-dimensional RG equations without changing the low energy Hamiltonian (1).

Assuming that the interactions are weak we linearize the spectrum close to the Fermi points:

$$c = R\mathrm{e}^{-\mathrm{i}k_F x} + L\mathrm{e}^{\mathrm{i}k_F x}, \quad f = r\mathrm{e}^{\mathrm{i}k_F x} + l\mathrm{e}^{-\mathrm{i}k_F x}, \quad (2)$$

where k_F is the Fermi momentum at $\mu = 0$. The kinetic energy acquires the standard form (we neglect the difference in the Fermi velocities of electrons and holes):

$$T = \mathrm{i}v(-R_\sigma^+ \partial_x R_\sigma + L_\sigma^+ \partial_x L_\sigma) + \mathrm{i}v(-r_\sigma^+ \partial_x r_\sigma + l_\sigma^+ \partial_x l_\sigma) \quad (3)$$

The Umklapp interaction becomes

$$V_2 = u_2 \Big\{ \Big[L_\uparrow L_\downarrow r_\downarrow^+ r_\uparrow^+ + R_\uparrow R_\downarrow l_\downarrow^+ l_\uparrow^+ +$$
$$\Big(R_\uparrow L_\downarrow + L_\uparrow R_\downarrow \Big) \Big(l_\downarrow^+ r_\uparrow^+ + r_\downarrow^+ l_\uparrow^+ \Big) \Big] + h.c. \Big\} \quad (4)$$

The densities are

$$n_c = (R_\sigma^+ R_\sigma + L_\sigma^+ L_\sigma) + \mathrm{e}^{-2\mathrm{i}k_F x} L_\sigma^+ R_\sigma + \mathrm{e}^{2\mathrm{i}k_F x} R_\sigma^+ L_\sigma$$
$$n_f = (r_\sigma^+ r_\sigma + l_\sigma^+ l_\sigma) + \mathrm{e}^{-2k_F \mathrm{i}x} r_\sigma^+ l_\sigma + \mathrm{e}^{2\mathrm{i}k_F x} l_\sigma^+ r_\sigma \quad (5)$$

The bosonization rules are standard:

$$L_\sigma = \frac{\xi_\sigma^{(1)}}{\sqrt{\pi a_0}} \mathrm{e}^{\mathrm{i}\sqrt{4\pi}\varphi_\sigma^{(1)}}, \quad R_\sigma = \frac{\xi_\sigma^{(1)}}{\sqrt{\pi a_0}} \mathrm{e}^{-\mathrm{i}\sqrt{4\pi}\bar{\varphi}_\sigma^{(1)}} \quad (6)$$

and the same for r, l with index 1 being replaced with 2. Here $\varphi, \bar{\varphi}$ are chiral bosonic fields and ξ are coordinate independent Klein factors:

$$\{\xi_\sigma^a, \xi_{\sigma'}^b\} = \delta_{\sigma\sigma'}\delta_{ab} \quad (7)$$

The Klein factors constitute the Clifford algebra and are Dirac γ-matrices of the O(6) group.

Substituting (6) into (4) and introducing new fields

$$\phi_c^{(+)} = \frac{1}{2}\Big[\phi_\uparrow^{(1)} + \phi_\downarrow^{(1)} + \phi_\uparrow^{(2)} + \phi_\downarrow^{(2)}\Big]$$
$$\phi_c^{(-)} = \frac{1}{2}\Big[\phi_\uparrow^{(1)} + \phi_\downarrow^{(1)} - \phi_\uparrow^{(2)} - \phi_\downarrow^{(2)}\Big]$$
$$\phi_s^{(+)} = \frac{1}{2}\Big[\phi_\uparrow^{(1)} - \phi_\downarrow^{(1)} + \phi_\uparrow^{(2)} - \phi_\downarrow^{(2)}\Big]$$
$$\phi_s^{(-)} = \frac{1}{2}\Big[\phi_\uparrow^{(1)} - \phi_\downarrow^{(1)} - \phi_\uparrow^{(2)} + \phi_\downarrow^{(2)}\Big] \quad (8)$$

we get

$$V = 4u_2 \frac{\xi_\uparrow^{(1)} \xi_\downarrow^{(1)} \xi_\downarrow^{(2)} \xi_\uparrow^{(2)}}{(\pi a_0)^2} \times \quad (9)$$
$$\cos[\sqrt{4\pi}\theta_c^{(-)}] \Big\{ \cos[\sqrt{4\pi}\phi_c^{(+)}] + \cos[\sqrt{4\pi}\phi_s^{(+)}] + \cos[\sqrt{4\pi}\phi_s^{(-)}] \Big\},$$

where $\phi = \varphi + \bar{\varphi}, \quad \theta = \varphi - \bar{\varphi}$. This interaction can be refermionized:

$$V = u_2 \bar{\psi}_0 \psi_0 \sum_{a=1}^{3} (\bar{\psi}_a \psi_a) \quad (10)$$

where $\bar{\psi}_a = (L^+, R^+)_a, \psi_a^T = (R, L)_a$. The new fermions transform in the vector representation of SO(8) group and should not be confused with original ones. The density-density interaction gives rise to the term

$$V_0 = -u_0 \sum_{a \neq b} (\bar{\psi}_a \psi_a)(\bar{\psi}_b \psi_b) \quad (11)$$

As a result we get the Gross-Neveu model with $Z_2 \times$O(6) symmetry:

$$\mathcal{H} = \mathrm{i}v \sum_{a=1}^{3} (-R_a^+ \partial_x R_a + L_a^+ \partial_x L_a) +$$
$$\mathrm{i}v(-R_0^+ \partial_x R_0 + L_0^+ \partial_x L_0) +$$
$$-u_0 \sum_{a \neq b} (R_a^+ L_a + L_a^+ R_a)(R_b^+ L_b + L_b^+ R_b) -$$
$$u_0' (R_a^+ L_a + L_a^+ R_a)^2 +$$
$$u_2 (R_0^+ L_0 + L_0^+ R_0) \sum_{a=1}^{3} (R_a^+ L_a + L_a^+ R_a)$$
$$-\mu (R_1^+ R_1 + L_1^+ L_1) - h(R_2^+ R_2 + L_2^+ L_2) \quad (12)$$

(here the index 0 corresponds to $(c, -)$, 1 to $(c, +)$, 2 and 3 to $(s, \pm)$). The RG equations can be extracted from [1]:

$$\dot{u}_0 = -4u_0^2 - 2u_2^2, \quad \dot{u}_2 = -(u_0' + 5u_0)u_2,$$
$$\dot{u}_0' = -4u_0^2 - 2u_2^2 \quad (13)$$

Let us consider the undoped ($\mu = 0$) case first. Analysis of Eqs.(13) shows that the interactions scale to strong coupling under rather general conditions. At strong coupling the RG trajectories asymptotically restore the SO(8) symmetry $u_0 = u_0' = |u_2|$. The spectrum is gapful and at the SO(8) symmetric point is known exactly (one can find a detailed discussion in [1]). The fact that the spectrum is gapful is an indication that the undoped state is not a superconductor. In the ground state fields $\phi_c^{(+)}, \phi_s^{(\pm)}$ freeze at $\Phi = 0$, the field $\theta_c^{(-)}$ freezes at 0 or $\sqrt{\pi}/2$ depending on sign of u_2 There is no *local* order parameter corresponding to these field configurations. This situation is known as *d*-Mott insulator [1]. It is interesting that the coherent spin excitations (vector particles) are emitted at the Neel wave vector connecting the